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The impact of two computer-aided emotion regulation strategies on risky driving behaviors in a virtual environment

Jiakun Liu¹, Xingpeng Wang¹ and Lei Zhu^{1*}

Abstract

Computer technology has been increasingly used to enhance driving safety, however, as a risk factor for driving safety, how anger can be effectively controlled by computer technology remains to be explored. Thus, the present study tested the impact of two computer-aided emotion regulation strategies (expressive suppression vs. cognitive reappraisal) on anger and risky driving behaviors in a virtual environment. After anger was triggered, participants were instructed by pre-set computer recording segments to use the above two strategies for emotion regulation. During driving, their facial expressions were recorded by a camera. It was found that only computer-aided expressive suppression could effectively regulate anger and reduce risky driving behaviors (i.e., angular overspeed and lateral acceleration overspeed). These findings suggest that the computer-aided expressive suppression has the potential to reduce risky driving behaviors for Chinese drivers and offer insights for the development of computer-aided approaches.

Keywords Emotion regulation, Risky driving behaviors, Virtual environment, Expressive suppression, Cognitive reappraisal

Background

With the rapid development of science and technology, computer technology has shown unprecedented potential in various areas. This progress has changed our lifestyles and work practices, introducing novel and innovative solutions. Computer technology has contributed significantly to the safety of drivers [1, 2], which has led to the development of different levels of autonomous driving. However, computer technology has led to a slight decline in road traffic fatalities in recent years, the World Health Organization has highlighted that road

traffic accidents persist as a global health crisis [3]. Considering that human-driven cars will continue to be the primary road users in the future, it is crucial to undertake comprehensive research on decreasing drivers' risky driving behaviors through computer aids. Existing studies have identified anger as a major factor in risky driving [4–6]. Therefore, how to effectively reduce anger through computer-aided emotion regulation in risky driving has become an urgent and important research question. An in-depth exploration of this issue will not only enhance our understanding of driver emotion regulation but also offer valuable references and guidance for future developments in computer-aided driving.

The effect of anger on risky driving behaviors

A widely shared viewpoint acknowledges the pivotal role that emotions play in shaping our everyday experiences

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[7–13]. Humans have six basic emotions: happiness, anger, sadness, fear, disgust, and surprise [14]. Subsequent studies refined these basic emotions [15, 16], revealing cross-cultural consistency in the expressions of emotions [17, 18]. Remarkably, anger, characterized by its extreme negative valence and high arousal [19], has attracted substantial scholarly attention [20, 21].

Many studies revealed that anger exerted a noticeable impact on individuals’ behavioral choices, probably resulting in undesired or adverse outcomes. In a study of social judgment, it was found that activating anger increased negative evaluations of ambiguous targets [22]. Likewise, within the field of work, anger has been linked to a reduction in overall work productivity [23]. More importantly, anger is often associated with risky behavior. When anger was induced, people reduced their evaluation of terrorism-related risk and showed a preference for more punitive measures [24]. Meanwhile, due to a false assessment of risk, angry people showed more risk-taking and reckless behaviors [11, 25].

Of note, there was considerable attention in numerous studies on the significant influence of emotions on driving [26], and particularly on the impact of anger [6, 27–30]. One line of research based on questionnaire studies indicated that anger in drivers was correlated with numerous risky driving behaviors [31–33]. Specifically, angry drivers might perceive a lower level of risk [31, 33], leading to poor driving behaviors [34], such as speeding, which can result in accidents [22, 35, 36]. Another line of research, particularly experimental studies, focused on the negative impact of anger on driving behaviors and responses, highlighting both operational and cognitive impairments associated with heightened anger levels. Deffenbacher et al. [37] reported that individuals experiencing anger exhibited a higher frequency of collisions. Moreover, they had an increased tendency to express anger through hostile and aggressive behaviors. Similarly, Abdu et al. [27] observed an increase in traffic violations and accelerated driving following anger induction. When trailing another vehicle, angry drivers showed reduced time for collision anticipation and an extended brake engagement period [6]. In addition, anger significantly reduced situational awareness and thus negatively affected the perception of

road traffic conditions [4]. In conclusion, anger is a significant threat to driving safety. It is essential to adopt effective approaches to managing and regulating drivers’ anger. However, effective regulation of anger in driving first requires its accurate identification. Facial expression recognition provides a promising approach to detecting anger through observable facial muscle movements.

Facial expression recognition

Facial expression recognition is one of the commonly employed techniques for identifying emotions, as facial expressions serve as direct indicators of affective states [38]. Among the various facial recognition approaches, the Facial Action Coding System (FACS) provides a comprehensive framework for analyzing anger-related facial muscle movements [39, 40]. Based on the anatomical characteristics of the face, FACS divides facial movements into several independent and interrelated Action Units (AUs). It systematically analyzes the relationship between these motion units and various emotional expressions [39, 40]. Basic emotions can be characterized by one AU or a combination of AUs. Anger is usually associated with a characteristic pattern of facial muscle activation, including AU4, AU5, AU7, and AU23 (Table 1).

Facial expression recognition provides valuable insights into individuals’ emotional states by identifying distinct facial muscle activations associated with specific emotions. Among these, anger is particularly relevant in various contexts, especially in driving behavior [27–30]. However, simply recognizing emotions is not sufficient—how individuals regulate their emotions after detection plays a critical role in ensuring driving safety.

Emotion regulation strategies

Referring to the Gross [42]’s definition, our study viewed emotion regulation as a systematic process by which individuals’ control and change their own or others’ emotions. In this process, emotions undergo specific changes in individual feelings, behaviors, and physiological responses through different emotion regulation strategies and mechanisms. The emotion regulation model includes two emotion regulation strategies [42]: reason-focused strategies and response-focused strategies. Specifically speaking, reason-focused strategies involve processing and adjusting the cause of the emotion. Reappraisal, a specific reason-focused strategy, involves rethinking the possible cause of the emotions to reduce the impact of stimulus-induced emotions. Response-focused strategies occur after emotion arousal, involving the individual’s processing and adjustment of the emotion already generated by their subjective experience. This includes expressive suppression, aiming at inhibiting facial expression [43]. Existing findings showed that cognitive reappraisal

Table 1 Anger-related action units [39, 41]

Action Unit (AU)	Facial Muscle Involved	Anger-related Expression
AU4	Corrugator supercilii, Depressor supercilii, Procerus	Brow lowering
AU5	Levator palpebrae superioris	Upper lid raising
AU7	Orbicularis oculi	Eyelid tightening
AU23	Orbicularis oris	Lip tightening

reduced emotional intensity by rethinking [44], whereas expressive suppression reduced emotional intensity by restraining expressions [45], both of which reduced the effects of stimulus-induced emotion production, called the regulatory effect [46]. A series of experimental studies showed that cognitive reappraisal could alter the intensity of emotions and even change emotions [47–50]. It achieved positive and healthy outcomes without increasing cognitive costs [42, 50]. Meanwhile, previous studies also explored the effectiveness of expressive suppression and found it beneficial in social interactions [51–53]. It has been found to be useful in modulating emotional response processing [54, 55]. It is worth noting that the function of expressive suppression was closely tied to cultural contexts [56–58]. Expressive suppression plays a positive role in collectivistic settings [59–61].

Importantly, in the area of driving, studies have emphasized the significance of emotion regulation. Specifically, difficulties in managing emotions have been associated with increased risky driving [28, 62], whereas enhanced emotion regulation ability has been associated with a decrease in aggressive behaviors [63]. In addition, risky driving behaviors might also be swayed by individual emotional personality traits such as spatial anxiety [9, 64, 65], which further emphasizes the need to explore how to regulate drivers' scene-induced or trait-related emotions by specific strategies.

However, there is still a gap in understanding how to reduce anger (i.e., road rage) during risky driving, since only a few studies using questionnaires focused on the effectiveness of different emotion regulation strategies during driving. Kwon & Oh [66] used online questionnaires to measure participants' preferences for emotion regulation strategies, emotions, and coping behaviors in response to driving stress. Structural equation modeling showed that negative emotions were effectively regulated through cognitive reappraisal. Similarly, Holman & Popușoi [67]'s study used questionnaires to measure driving styles, emotions, and the habitual use of emotion regulation strategies. Mediation analyses revealed that cognitive reappraisal and expressive suppression to attenuate the intensity of anger had a positive effect on driving behaviors. These studies suggested that instructing drivers to use cognitive reappraisal and expressive suppression when they feel anger might be an effective method. However, the above two studies were based on self-reflection which had internal bias and low ecological validity. Measurements of driving behaviors might better illustrate how different strategies regulate emotional responses during driving.

More importantly, when drivers encounter unexpected situations that trigger anger while driving, it is essential to effectively assist them in regulating their emotions in a timely and externally focused manner to

reduce dangerous driving. This involves employing particular emotion regulation strategies. While adaptive music, ambient lighting, and other methods can assist in regulating drivers' emotions, their effectiveness varies significantly due to individual differences, making precise interventions for anger particularly challenging. For instance, the relaxing effect of music depends on personal preferences [68]. In contrast, cognitive reappraisal and expressive suppression allow for targeted interventions based on a driver's current emotional state. For instance, cognitive reappraisal helps drivers reinterpret [42, 44], while expressive suppression requires them to actively control their facial expressions and body movements [43, 45], thereby regulating anger. Moreover, as driving requires maintaining concentration while managing emotions, music-based relaxation techniques may divert attention [69, 70], while ambient lighting provides only indirect regulation and may be insufficient for alleviating intense driving-related anger [71]. Consequently, compared to adaptive music and ambient lighting, cognitive reappraisal and expressive suppression might offer more direct and precise emotional regulation, facilitating rapid recovery of composure and reducing the likelihood of anger-induced risky driving behaviors. Notably, Gross [42] identified cognitive reappraisal and expressive suppression as two of the widely adopted and effective strategies.

There were several previous studies providing the manipulations of cognitive reappraisal and expressive suppression (Table 2). In the case of cognitive reappraisal, researchers more often encouraged participants to consider the reasons for the events from an objective perspective and provide reasonable explanations. As for expressive suppression, it involved suppressing facial expressions to prevent others from perceiving emotions.

In addition, computer-aided driving has been extensively integrated into the driving process [77, 78] and can effectively reduce the occurrence of accidents [2]. For instance, a previous study found that the augmented reality map and the map navigation played complementary roles in helping to better achieve driving goals and ensure safety [79]. Furthermore, computer-aided interventions improved drivers' skills in the face of extreme situations and reduced driving risks [1, 2]. Considering that driving is a highly dynamic and cognitively demanding task. Traditional emotion regulation methods, such as deep breathing or meditation, require active effort from the driver [30]. However, drivers often cannot focus on managing their emotions while driving. In contrast, computer-aided emotion regulation strategies might offer a timely and proactive approach to managing emotions, thereby enhancing both emotional stability and driving safety. Meanwhile, with the computer-aided technology, the uniformly set regulatory guidance phrases can be

Table 2 The manipulations of emotion regulation strategies

Reference	Cognitive Reappraisal	Expressive Suppression	Presentation
Franz et al. [72]	Encourage participants to think that things will go better or that things are fake or made up.	Encourage participants to not reveal your feelings	Instruction
Z. Liu et al. [73]	Encourage participants to reconsider negative aspects from a positive perspective.	Encourage participants to calm down and hide their negative emotions.	Each strategy offers three examples for participants to learn.
Livingstone & Isaacowitz [74]	Encourage participants to take a detached and calm attitude or consider it positive aspect to minimize your negative emotions.	Encourage participants to not show emotions on their face so people can't see how they feel.	Instruction
Mohammed et al. [75]	Encourage participants to objectively consider the negative pictures and remain calm.	Encourage participants to hide their feelings about the negative pictures.	Instruction
Nook et al. [76]	Encourage participants to create a story for each image to make it less offensive	/	Instruction
Zhu et al. [46]	Tell participants that the emotions of the pictures were unrelated to personal emotions.	Encourage participants to maintain a neutral facial expression.	Instruction

preloaded onto the device, thereby forming a standardized guidance process and providing support for the optimization of the future system. However, the effectiveness of emotion regulation through computer techniques has not been investigated.

The current study

Thus, the current study aimed to investigate the effects of two computer-aided emotion regulation strategies on risky driving behaviors. Considering the cost of studying driving safety in the real world and the effectiveness of studying driving behaviors in a virtual environment [1, 80–82], the experiment was carried out in a virtual environment. In the experiment, the participants were required to drive in virtual scenes after inducing anger and after emotion regulation. Two kinds of computer-aided emotion regulation instructions (cognitive reappraisal and expressive suppression) were given auditorily to two emotion regulation groups through the earphones. For comparison, a control group without any emotion regulation manipulation was also enrolled. Risky driving behaviors were indexed by the driving speed,

acceleration speed, angular speed and lateral acceleration speed during driving. The index values before and after emotion regulation were recorded to examine the effect of two computer-aided emotion regulation strategies. It was hypothesized that the risky driving behaviors would decrease after emotion regulation compared to the control group. Considering that it was not convenient to give ratings while driving, we only required the participants to rate their emotions after driving. Instead, to examine the effectiveness of two emotion regulation strategies, participants' facial expressions were also recorded while driving by a camera.

Method
Design

A 3 emotional regulation group (expressive suppression, cognitive reappraisal and control) × 2 block (after emotion initiation, after emotion regulation) mixed design was adopted with the emotion regulation as a between-subject variable and the block as a within-subject variable.

Apparatus

A Logitech G29 driving simulator, featuring a force-feedback steering wheel and integrated throttle and brake pedals, was employed. The experimental program was run on an Alienware desktop computer and presented on an AOC AH-IPS 23.8-inch monitor with a resolution of 1920 × 1080 pixels and a refresh frequency of 60 Hz. To record participants' facial expressions, a front screen was used to present the driving scene rather than the virtual reality headset. Additionally, a Logitech C920e camera, with a resolution of 640 × 480 pixels and a frame rate of 10 frames per second, was placed at the top of the screen to capture the participants' facial expressions while driving (Fig. 1).

Materials

Unreal Engine 4 based on the dynamic modeling by Zhou and Zheng [83] was used to build the experimental scenes. To reduce the practice effect and fatigue effect, as well as to keep the road type and driving context constant, three driving scenes containing different details were produced: countryside, city, and highway, using buildings, landscapes, and traffic signs to represent differences in the scenes. The scenes were presented in a fixed order: countryside followed by city and finally highway, without extreme weather. Additionally, traffic jams were not simulated and no other cars were in front of the participant's car except in the phase of emotion initiation. The total driving length was more than 7 km, with a 40 km/h speed limit to reflect a common urban speed limit. All the roads consisted of four lanes and were bidirectional, except for one kilometer, where the



Fig. 1 The driving environment settings

roads expanded to six lanes. The entire driving process required seven turns and took approximately 10 min.

In order to counterbalance driving experience and some personality related to risky driving between the three groups, before the experiment, participants needed to complete the Sensation Seeking Inventory ($\alpha=0.87$; Chinese version by Wang et al., 2000), the Trait Anger Scale ($\alpha=0.78$; Chinese version by Liu & Gao, 2012) which have been used and validated in previous studies [84–86]. Additionally, participants needed to complete a self-designed Driving Behavior Questionnaire.

Procedure

Before the experiment, all the participants were required to complete the Sensation Seeking Inventory [85], the Trait Anger Scale [84, 86] and the Driving Behavior Questionnaire. Then, they were taken to a separate room with blackout curtains to ensure consistent lighting. In this room, they sat comfortably in front of the computer and driving simulator. They completed a 5-minute training session to become familiar with the driving operations. After practice, they proceeded to the formal experiment with two block periods: after emotion initiation and after emotion regulation.

Emotion initiation Referring to previous studies [5, 87], anger was induced by the interaction in traffic events. Specifically, Cai & Lin [87] adopted the interactional car to cut in and brake, while Roidl et al. [5] chose the braking cars. Additionally, recalling and watching traffic clips demonstrated that intentional overtaking and abrupt braking usually induce anger in drivers [35, 87, 88]. Thus, this study set up overtaking and braking scenarios with the following specific procedures:

Before the first period, the participant's car was instructed to drive in the left lane. During the drive, four cars sequentially overtook from the right lane, each merging into the left lane and then abruptly braking in front of the participant's car (Fig. 2). The low traffic density and lack of traffic jams indicated that the overtaking and abrupt braking were intentional actions by other cars. After that, there were no further requirements about which lane the participants should keep in and the participants drove about 3.4 km with 4 turns. The driving behaviors were recorded every 10 m.

Emotion regulation Considering the effectiveness of voice-guided emotional regulation [72, 75, 76, 89], before the second period, computer-aided emotion regulation strategies were given auditorily in a female voice twice through the earphones. Then, the participants were required to repeat the strategy instruction silently three times. The whole emotion regulation period lasted for about 1 min and 40 s including auditory instruction, silent repeating and its related emotional adjustments. For the expressive suppression group, the participants were instructed "While feeling your emotions, do not show your feelings, try to hide your expression, and do not let others see your facial expressions". For the cognitive reappraisal group, the participants were instructed "It is normal to be overtaken during driving, and the car overtaking you is carrying some emergency patients. Please think about the overtaking behaviors with an objective and rational attitude" (Fig. 2). However, the control group experienced a passive waiting period equivalent to the experimental groups, but received no instructions after being overtaken. Then, the participants

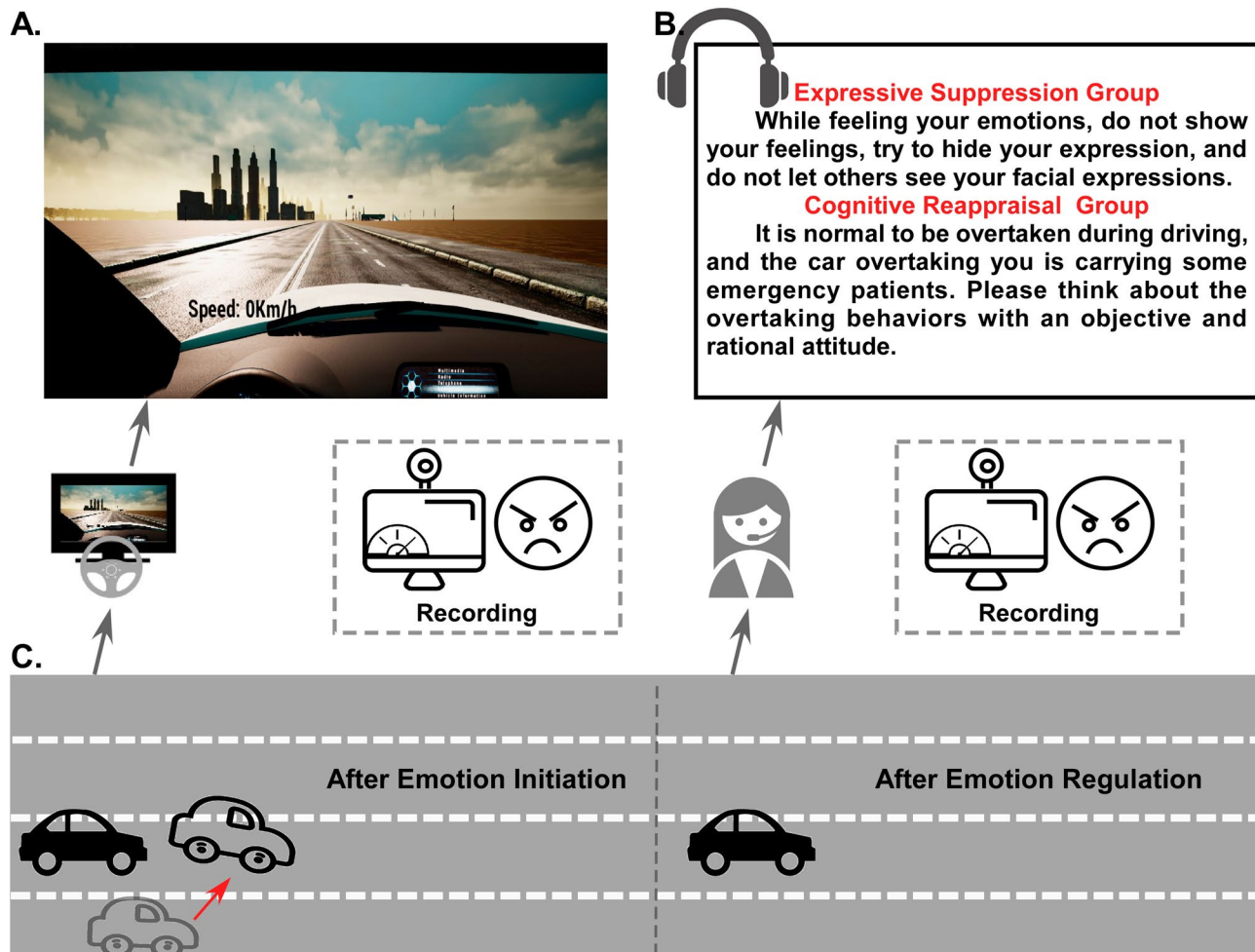


Fig. 2 Experimental procedure. (A) Driving scene. (B) Emotion regulation instructions. (C) Two blocks

drove about another 3.4 km with 3 turns. The driving behaviors were recorded every 10 m.

Finally, the participants were required to rate their basic emotions (happy, angry, nervous, fearful, and neutral) on a 7-point Likert scale (1 indicating extremely unhappy, etc., 7 indicating extremely happy, etc.). During two blocks, the participants' risk-driving behaviors were recorded and their facial expressions were recorded by a camera.

Participants

A priori power analysis (G*Power 3.1) with an effect size of 0.25, an alpha level of 0.05, and a desired power ($1 - \beta$) of 0.95, focusing on the interaction between 3 emotion regulation (expressive suppression, cognitive reappraisal and control) and 2 block (after emotion initiation, after emotion regulation) in a repeated-measure ANOVA, indicated a sample size of 66. The study recruited 66 drivers (39 females, $M_{\text{age}} = 22.11$, $SD = 3.15$) from the university community with several requirements: (1) Have a Category C driver's license (authorized for driving small passenger vehicles with no more than 9 seats), (2) Have driven a car at least once within the past 6 months, (3) Have a normal or corrected-to-normal vision, (4) Have no motion sickness or Virtual Reality (VR) discomfort. They were randomly assigned to three groups: the control group ($N = 22$), the cognitive reappraisal group ($N = 23$), and the expressive suppression group ($N = 21$). All of them reported no mental illness or psychological disorders. Additionally, they reported no history of central nervous system surgery or physical disabilities.

Data analysis

The present study used the speed, acceleration speed, angular speed and lateral acceleration speed as indices for risky driving behaviors. To obtain the representative indicators of driving behaviors, the driving data recorded throughout the driving task were aggregated by calculating their mean values. Both the large speed and the large acceleration speed was associated with an increased risk of traffic accidents [5, 90, 91]. Both the forward and the backward acceleration speed, as well as the leftward and the rightward lateral acceleration speed, were recorded as positive values. In other words, the large lateral acceleration speed, regardless of direction, was considered risky. Finally, the large angular speed indicated that drivers were more inclined to make sharp turns with a small radius [92, 93], which easily led to loss of control.

Besides, considering that the incidence and severity of accidents on bending roads are usually higher than on straights [94–97], the driving behaviors on the straight and bending roads were analyzed separately. The bending roads were defined as 25 m before and after the turns. The speed was taken as an index for risky driving

behaviors on straight roads. The angular speed was taken as an index for risky driving behaviors on bending roads. The acceleration and lateral acceleration speeds were taken as indices for risky driving behaviors on both the straight and bending roads.

In addition, for the facial expression videos, the machine learning software OpenFace 2.2, which has shown its applicability to the Chinese population [98, 99], was used to recognize and detect facial behaviors [100]. The software extracts the mean and maximum activation intensity of each facial Action Unit (AU), ranging from 0 (lowest intensity) to 5 (highest intensity). The mean and maximum intensity of AU4, AU5, AU7, and AU23 were of interest since they were associated with anger expressions [15, 101, 102]. Finally, repeated-measures ANOVAs were conducted on emotional indicators and risky driving behaviors to examine the different effect of the emotion regulation strategies. To further compare the effects of the emotion regulation strategies, the change scores (after emotion induction minus after emotion regulation) were calculated, and a one-way ANOVA was performed on these change scores.

Moreover, to test the relation between facial expressions and driving behaviors, the Pearson correlation analyses were carried out. Since there are significant individual differences in emotional expression [103–107], to eliminate the effect of individual differences on the correlations, we calculated the standardized changes for the facial expression indicators (mean and max intensities of AU5, AU7, and AU23) by the formula:

$$\frac{\text{after emotion initiation} - \text{after emotion regulation}}{\text{after emotion initiation}}$$

Also, the standardized changes were calculated for the driving behavior indicators (angular speed and lateral acceleration speed). Then, the correlations between facial expression changes and risky driving behavior changes were analyzed to examine whether reductions in anger were associated with decreases in risky driving behaviors.

Results

The pre-analysis of the three scales data was conducted. Specifically, the Sensation Seeking Inventory ($\alpha = 0.87$) [85, 108, 109] and the Trait Anger Scale ($\alpha = 0.78$) [84, 110] demonstrated high reliability. There were no significant differences between the three groups in Sensation Seeking, Trait Anger and driving experience.

Emotions

A 3 emotional regulation group (expressive suppression, cognitive reappraisal and control) $\times$ 2 block (after emotion initiation, after emotion regulation)

repeated-measure ANOVA on the max intensity of AU5 indicated that the interaction between block and emotional regulation group was significant ($F(2,63)=3.27$, $p=.045$, Fig. 3). Further analyses revealed that for the expressive suppression group, the max intensity of AU5 after emotion initiation was higher than that after emotion regulation, but this trend was marginally significant, $F(1,20)=3.57$, $p=.054$. However, the effects of block for the cognitive reappraisal ($F(1,22)=3.25$, $p=.085$) and control groups ($F(1,21)=1.38$, $p=.254$) reflected statistical inconclusiveness.

Then, a one-way ANOVA showed the main effect of the group on the intensity change of AU5 ($F(2,63)=3.27$, $p=.045$). Further analyses revealed that the intensity change of the expressive suppression group was significantly greater than that of the control group, $t(41)=2.34$, $p=.048$. However, the differences in the intensity change between the cognitive reappraisal group and the control group ($t(43)=2.07$, $p=.105$), and between the cognitive reappraisal group and the expressive suppression group ($t(42)=0.32$, $p=.945$) did not reach statistical significance.

Besides, we also calculated the Bayes factors (B) to assess the strength of evidence for the simple effects of block. A B value exceeding 3 strongly supports the alternative hypothesis and below 1/3 shows substantial evidence for the null rather than the alternative hypothesis, while values between 3 and 1/3 suggest data insensitivity, indicating unclear evidence for either the alternative or null hypothesis [111, 112]. For the expressive suppression group, B was 4.30, strongly supporting that the max intensity of AU5 after emotion initiation was higher than that after emotion regulation. However, B was 2.55 for the cognitive reappraisal group, indicating that the evidence was inconclusive for either the alternative or the null hypothesis. For the control group, B was 0.13, strongly supporting that the effect of block was not significant. The findings indicated that at least the manipulation of expressive suppression was successful.

Additionally, the repeated-measure ANOVAs on the mean and maximum activation intensity of AUs showed that the mean intensity of AU23 was higher after emotion initiation than that after emotion regulation ($F(1,63)=4.22$, $p=.044$). Besides, the max intensity of AU7 and AU23 was higher after emotion initiation than that after emotion regulation (AU7: $F(1,63)=14.08$, $p<.001$; AU23: $F(1,63)=5.34$, $p=.024$).

For the ratings of basic emotions (happy, angry, nervous, fearful and neutral), one-way ANOVAs indicated that the difference between different groups did not reach statistical conclusiveness. This might be due to the reason that the emotions induced by overtaking had already disappeared or been adjusted after about 10-minute driving.

Risky driving behaviors

For the bending roads, a 3 emotional regulation group (expressive suppression, cognitive reappraisal and control) $\times$ 2 block (after emotion initiation, after emotion regulation) repeated-measure ANOVA on angular speed indicated that the interaction effect between block and emotional regulation group was significant ($F(2, 63)=35.51$, $p<.001$, Fig. 3). Further analyses indicated that compared to after emotion initiation, a smaller angular speed was observed for the expressive suppression group after emotion regulation ($F(1, 20)=114.59$, $p<.001$). However, no statistically conclusive difference was observed for the other two groups.

Then, a one-way ANOVA showed a significant main effect of the group on the change of angular speed ($F(2, 63)=35.51$, $p<.001$). Further analyses revealed that the angular speed change was significantly larger for the expressive suppression group than for the cognitive reappraisal group ($t(42)=7.81$, $p<.001$) and for the control group ($t(41)=6.76$, $p<.001$). However, the difference of the angular speed change between the cognitive reappraisal group and the control group ($t(43)=-0.10$, $p=.586$) reflected statistical inconclusiveness. Moreover, the main effect of block on angular speed was significant ($F(1, 63)=57.17$, $p<.001$). Compared to after emotion initiation, a smaller angular speed was observed after emotion regulation.

A repeated-measure ANOVA on lateral acceleration speed showed that the interaction effect between block and emotional regulation was significant ($F(2, 63)=19.90$, $p<.001$, Fig. 3). Further analyses revealed that compared to after emotion initiation, a smaller lateral acceleration speed was observed for the expressive suppression group after emotion regulation ($F(1, 20)=68.05$, $p<.001$, $\eta_p^2=0.77$). However, no statistically conclusive difference was observed for the other two groups.

Then, a one-way ANOVA showed a significant main effect of the group on the change of lateral acceleration speed ($F(2, 63)=19.90$, $p<.001$). Further analyses revealed that the lateral acceleration speed change for the expressive suppression group was significantly greater than that for the cognitive reappraisal group ($t(42)=6.04$, $p<.001$) and for the control group ($t(41)=4.68$, $p<.001$). However, the difference of the lateral acceleration speed change between the cognitive reappraisal group and the control group ($t(43)=-1.33$, $p=.386$) reflected statistical inconclusiveness. Moreover, the main effect of block was significant on lateral acceleration speed ($F(1, 63)=21.53$, $p<.001$). Compared to after emotion initiation, a smaller lateral acceleration speed was observed after emotion regulation. For the acceleration speed, none of the main effects and interaction reached statistical conclusiveness. The details of the significant results, including effect sizes and confidence intervals, can be found in Table 3.

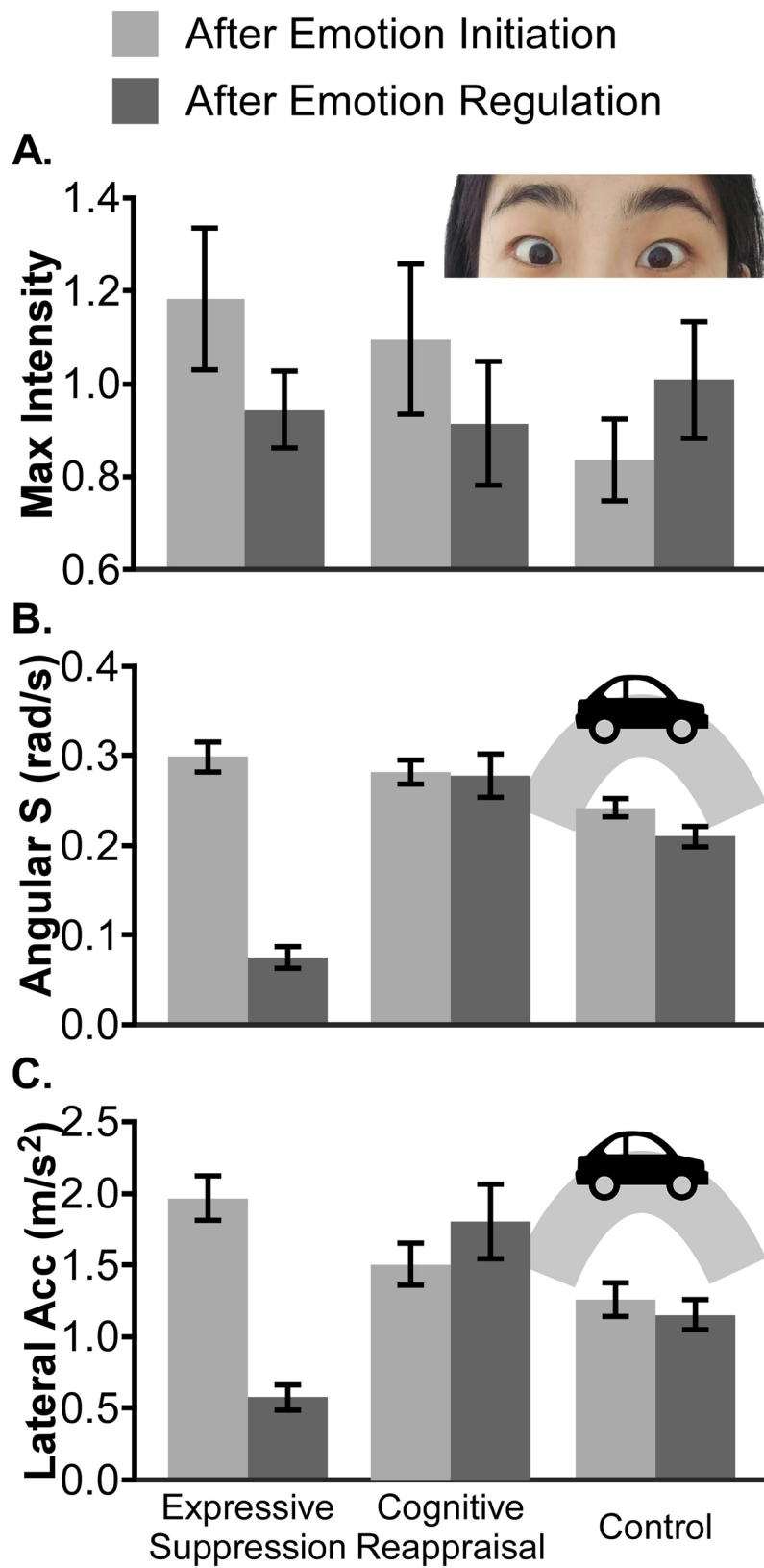


Fig. 3 Results of the facial expression and the risky driving behaviors for three groups. **(A)** The max activation intensity of AU5 for three groups. **(B)** The angular speed at bending roads for three groups. **(C)** The lateral acceleration speed at bending roads for three groups. Lateral acc refers to the lateral acceleration speed. Angular S refers to the angular speed. Error bars indicate $\pm$ standard error

Table 3 The summary of significant results on emotions and risky driving behaviors

Dependent Variable	Effects	F or t	p	Effect Size	90% CI or 95% CI
Emotions					
AU5 _{Max}	emotional regulation group × block	3.27	0.045	0.09	[0.01, 0.21]
	block _{expressive suppression}	3.57	0.054	0.17	[0.12, 0.35]
AU5 _{Change}	emotional regulation group	3.27	0.045	0.09	[0.01, 0.21]
	expressive suppression vs. control	2.34	0.048	0.66	[0.03, 1.30]
AU7 _{Max}	block	14.08	< 0.001	0.18	[0.06, 0.32]
AU23 _{Mean}	block	4.22	0.044	0.06	[0.01, 0.18]
AU23 _{Max}	block	5.34	0.024	0.08	[0.01, 0.20]
Risky Driving Behaviors					
Angular speed	emotional regulation group × block	35.51	< 0.001	0.53	[0.38, 0.63]
	block _{expressive suppression}	114.59	< 0.001	0.85	[0.18, 0.27]
	block	57.17	< 0.001	0.48	[0.33, 0.59]
Angular speed _{Change}	emotional regulation group	35.51	< 0.001	0.53	[0.38, 0.63]
	expressive suppression vs. cognitive reappraisal	7.81	< 0.001	2.16	[1.40, 2.90]
	expressive suppression vs. control	6.76	< 0.001	2.26	[1.48, 3.02]
Lateral acceleration speed	emotional regulation group × block	19.90	< 0.001	0.39	[0.23, 0.51]
	block _{expressive suppression}	68.05	< 0.001	0.77	[0.59, 0.85]
	block	21.53	< 0.001	0.25	[0.11, 0.39]
Lateral acceleration speed _{Change}	emotional regulation group	19.90	< 0.001	0.39	[0.23, 0.51]
	expressive suppression vs. cognitive reappraisal	6.04	< 0.001	1.64	[0.95, 2.32]
	expressive suppression vs. control	4.68	< 0.001	1.72	[1.01, 2.42]

For the straight roads, the 3 × 2 repeated-measure ANOVAs on speed, acceleration speed and lateral acceleration speed revealed that none of the main effects and the interactions reflected statistical conclusiveness.

Lastly, since only the expressive suppression group showed a significant emotional regulation effect, the correlation analyses were conducted on the standardized changes of facial expression indicators (mean and max intensities of AU5, AU7, AU23) and driving behavior indicators (angular speed and lateral acceleration speed) separately for the three groups. For the expressive suppression group, the correlation between the max intensity of AU23 and angular speed was marginally significant ($r = .42$, $p = .060$). Results from the other groups were statistically insensitive, indicating a lack of conclusive evidence. Similarly, the correlation between max intensity of AU23 and lateral acceleration speed was only marginally significant ($r = .38$, $p = .089$) for the expressive suppression group. Results from the other groups were statistically insensitive, indicating a lack of conclusive evidence. Results of the mean intensities of AUs and risky behavior indicators were statistically insensitive, indicating a lack of conclusive evidence.

Discussion

The current study aimed to investigate the effects of two computer-aided emotion regulation strategies (expressive suppression and cognitive reappraisal) on risky driving behaviors. The results showed that only the computer-aided expressive suppression strategy decreased risky

driving behaviors. Several aspects of the main results were discussed.

The successful manipulation of anger

First, it was difficult for the participant to give their emotion ratings while driving. So, they were required to rate basic emotions (happy, angry, nervous, fearful and neutral) after driving and there was no statistical evidence of any group differences in ratings. This might be due to the reason that the emotions induced by overtaking had already disappeared or been adjusted after about 10-minute driving. However, the facial expression data might also give insights. The mean and max intensity of the AUs after emotion initiation were higher than those after emotion regulation. Combined with previous demonstration that overtaking and braking commonly induce anger [5, 87, 88], these findings plausibly indicated successful anger induction.

Given that anger can significantly affect driving safety and lead to aggressive behavior [27, 31, 32, 37], it is particularly important to deeply understand how drivers can effectively manage their anger. Studies have shown that various factors, including an individual's personality traits [64, 113], driving habits [9, 22, 35, 36], and the environmental factors [94, 97], all impact the driver's ability to manage anger. For instance, higher levels of forgiveness are associated with adaptive reactions to anger [113]. In contrast, bad driving habits, such as frequently speeding, or using mobile phones, not only distract the driver's attention [22, 35] but also may intensify the occurrence of

anger [35, 36]. Therefore, in order to effectively manage anger, drivers might benefit from good driving habits and learn to apply effective emotional regulation strategies.

The effect of expressive suppression on risky driving behaviors

For risky driving indices, the results indicated that the angular speed and lateral acceleration speed significantly decreased after expressive suppression, indicating that drivers instructed by the computers to suppress their facial expressions could effectively reduce risky driving behaviors. However, the regulatory effect of cognitive reappraisal on risky driving behaviors was not satisfactory. The same results were found for the control group. Meanwhile, comparisons of the changes between after emotion initiation and after emotion regulation between the groups revealed that the expressive suppression group showed a significantly larger reduction than both the cognitive reappraisal and control groups. These results partially verified our hypothesis, highlighting the regulatory effect of the computer-aided expressive suppression on risky driving behaviors. Also, it was consistent with the previous study by questionnaires [67] demonstrating that expressive suppression was an effective strategy for risky driving.

The reasons for the effectiveness of expressive suppression

The effectiveness of expressive suppression might be attributed to its ability to reduce the processing of emotional stimuli within a short period during emotion regulation [114], reflecting its immediacy. During such a short period, compared to people from other countries, Chinese people might automatically select the expressive suppression strategy they often use [59–61], since it consumes fewer resources in complex driving environments.

Furthermore, Chinese preference for such strategies might be closely tied to Chinese culture. The strategy of expressive suppression might align with the social norm of enhancing group harmony embedded in collectivist societies. It was found that the effectiveness of expressive suppression was related to the specific culture where it was used [56]. It might be effective only in collectivist societies since individuals in collectivist societies were likely to sacrifice their benefits or suppress their willingness or thoughts to promote group harmony and coherence [115]. Correspondingly, to facilitate social adjustment, they might be inclined to use strategies such as expressive suppression to inhibit their true thoughts and try to follow the social norms [59, 60]. Previous studies have consistently shown that individuals from some collectivist cultures in Asia tend to employ expressive suppression more frequently, particularly in regulating negative emotions [56, 116]. Moreover, previous studies have demonstrated the positive effects of expressive

suppression on individuals in China (i.e., a typical collectivist society) in various aspects, including interpersonal relationships [51–53], job performance [61], and psychological well-being [53]. The present study further extended the positive effect of expressive suppression on controlling risky driving behaviors for Chinese drivers. And, more importantly, these findings suggest that the computer-aided expressive suppression might be an easy and effective way to reduce risky driving behaviors for Chinese drivers in their real life.

The reasons for the limited effectiveness of cognitive reappraisal

However, the present study found that the effect of cognitive reappraisal was inferior to expressive suppression, which was inconsistent with Kwon & Oh [66] and Holman & Popușoi [67]. This inconsistency might be due to the different methods used in the studies. Kwon & Oh [66] and Holman & Popușoi [67] used subjective ratings, whereas the present virtual driving study measured several risky driving behavioral indices. Previous studies showed that cognitive reappraisal mainly modulated semantic representations of an emotional stimulus using cognitive control [47], leading participants to rationalize their judgments cognitively. Thus, it might bring more effects on subjective ratings (cf [66, 67]), than behavioral responses. Further, subjective ratings that have self-reflection bias may have low ecological validity. The measurement of subjective ratings may lead participants to be aware of the purpose of the study and be inclined to give ratings that comply with the study's aim. That might also be the reason for the positive effect of cognitive reappraisal shown by Kwon & Oh [66] and Holman & Popușoi [67]. On the other hand, behavioral responses during virtual driving might not be easily controlled by one's consciousness and might be immune to the influence of cognitive reappraisal.

This consistency might also be due to whether the task had a time limit. The measurement of subjective ratings used by Kwon & Oh [66] and Holman & Popușoi [67] did not have a time limit. In contrast, the duration of cognitive reappraisal in the present study might not be sufficient for participants to fully process and reinterpret. Meanwhile, the complex virtual driving task used in this study demanded rapid responses from participants. As Hu et al. [10] emphasized, time pressure affected risk-taking behaviors. Under such time pressure, the potential effectiveness of cognitive reappraisal might have been limited. Moreover, Mohammed et al. [75] and Yuan et al. [117] found that cognitive reappraisal was slower in regulating negative emotions compared to expressive suppression. Additionally, neurophysiological evidence suggested that cognitive reappraisal was associated with executive function and required cognitive resources

[118]. Thus, the participants of the present study might not have enough time to reappraise the overtaking cognitively and thus adjust their risky driving behaviors.

Furthermore, the efficacy of cognitive reappraisal might vary depending on the specific type of emotion being regulated [119]. Anxiety, particularly social anxiety, is often characterized by heightened suppression of emotional expression, largely driven by concerns about potential negative outcomes [120]. Cognitive reappraisal could relieve such concerns by enabling individuals to reinterpret the emotional meaning of a situation [121]. From a neurobiological perspective, individuals with social anxiety frequently exhibited reduced activation in the frontoparietal network, which was crucial for effective emotion regulation [122, 123]. Since cognitive reappraisal relies on the engagement of these regions, its use might not only help reduce negative affect but also contribute to the strengthening of emotional regulation capacities over time. In contrast, anger is closely associated with impairments in inhibitory functions [124, 125]. Previous research has identified this specific deficit as fundamental to state anger [126]. Consequently, cognitive reappraisal may have limited efficacy in regulating anger, because the cognitive mechanisms underlying anger differ from those involved in anxiety regulation.

Although previous studies by questionnaires provided evidence that emotion regulation was important for risky driving [28, 62, 63], there is limited evidence for which computer-aided emotion regulation strategies effectively reduce risky driving behaviors. In this sense, our findings first support that computer-aided expressive suppression might be an effective strategy for reducing risky driving behaviors in a virtual driving environment, contributing to safe driving. Furthermore, our findings also contribute to enhancing computer-aided approaches to serving driving needs.

The effect of expressive suppression on anger

For facial expression, the results also suggested that expressive suppression effectively reduced the intensity of anger. First, for the expressive suppression group, a lower max intensity of AU5 after emotion regulation was observed compared to after emotion initiation. However, the regulatory effect of cognitive reappraisal on the max intensity of AU5 reflected statistical inconclusiveness. The same results were found for the control group. Regarding the changes between after emotion initiation and after emotion regulation, a significant larger reduction was observed for the expressive suppression group compared to the control group. According to the FACS, AU5, describing the upper lid raised, was commonly associated with anger [15, 101, 102]. These findings support the effectiveness of expressive suppression in regulating anger-related facial expressions. This is consistent

with previous research demonstrating that expressive suppression could effectively regulate negative emotions [54, 55].

Moreover, from the direction of AU5 intensity change, it can be found that the intensity of AU5 decreased in both the expressive suppression group and cognitive reappraisal group, although the cognitive reappraisal group demonstrated less favorable outcomes. In contrast, AU5 intensity increased for the control group, although the effect was insignificant, indicating that without active emotion regulation, anger-related facial expressions might intensify over time. This means that anger regulation is crucial. If anger is not effectively managed, it may accumulate, thereby affecting driving safety.

Additionally, the analysis revealed a correlation between the anger facial expression (max intensity of AU23) and the risky driving behavior, although it was only marginally significant. Similarly, the interaction effect on the max intensity of AU5 also demonstrated marginal significance. And, the correlation and interaction effect appeared in different AUs (AU23 vs. AU5). This might be attributed to the small sample size ($N=21$) after grouping, which limited statistical power and affected the significance testing of the correlations. Additionally, since driving behavior data were recorded at an interval of 10 m while facial expressions were captured as continuous real-time videos, it was difficult to precisely align the timing between facial expressions and driving behavior data. Due to this limitation, a more in-depth analysis of the dynamic relations between facial expressions and driving behaviors was constrained.

In summary, these results provide valuable insights into effective emotion regulation strategies for risky driving and suggest potential directions for developing computer-aided approaches to address drivers' emotional needs, although further research is needed to validate these findings in real-world environments. Meanwhile, computer-aided systems can uniformly integrate effective guidance information into the simulator, thereby constructing a standardized and repeatable intervention mechanism. This method provides a feasible reference path and theoretical support for the design and optimization of future ADAS. In the future, when a Chinese driver is detected showing anger, the system might instruct the driver to use expressive suppression strategies to help regulate his or her emotions, thereby reducing risky driving behaviors. However, the application still requires further empirical validation.

Limitations and future studies

There are still several limitations. First, the virtual driving environment with only straight and bending roads was relatively simple compared to real-life driving scenarios. Numerous studies have demonstrated that driving

scenarios greatly influence risky driving behavior [127, 128]. Future studies might simulate more realistic driving situations by introducing more different types of roads to investigate the effect of the computer-aided expressive suppression strategy. Meanwhile, due to the lack of somatosensory feedback in the virtual environment and other differences from the real driving scene, it might lead to a decrease in the feeling of immersion, thereby weakening the intensity of anger they might experience. Therefore, there are potential limitations in mimicking real-world driving conditions. To support broader application, future research might validate the regulatory effect of computer-aided expressive suppression in real-world driving environments. Second, although the results found the intensity of AUs was higher after emotion initiation than after emotion regulation, the effectiveness of anger initiation remains to be explored (cf [5]). Future studies might also measure anger by multiple indices such as measure the facial expression and skin conductance responses at the same time to confirm the successful manipulation of anger. Furthermore, the resolution and frame rate of the camera used might limit the detection of very brief or subtle facial muscle activations. To address this, future studies might explore the use of more refined measurement tools, such as facial electromyography (fEMG), or enhance imaging setups with higher-resolution and higher-frame-rate cameras. Thirdly, collecting self-reported emotions only after the experiment limited their effectiveness as a sensitive manipulation check for immediate emotional states following induction or regulation. The future research might collect self-reports at more temporally suitable and relevant points. Last but not least, this study focused solely on the effectiveness of emotional regulation during real-time driving. However, driving is a dynamic and interactive process, where previous driving experiences may influence subsequent ones [129]. The combination of immediate and long-term regulation should be considered. Future studies could employ a longitudinal tracking design to gain a more comprehensive understanding of the long-term effects of these strategies by continuously observing participants' emotional and risky driving behaviors.

Conclusions

In conclusion, the present study aimed to explore how different computer-aided emotion regulation strategies adjusted anger and risky driving behaviors in a virtual environment. It was revealed that, compared to cognitive reappraisal, the computer-aided expressive suppression was able to reduce risky driving behaviors and the intensity of anger for Chinese drivers. Our findings provide valuable insights into reducing drivers' risky driving behaviors in China.

Abbreviations

FACS	Facial Action Coding System
AUs	Action Units
AU	Action Unit
Lateral Acc	Lateral Acceleration Speed
Angular S	Angular Speed
ADSD	Advanced Driver Assistance Systems

Author contributions

Jiakun Liu: Formal analysis, Validation, Visualization, Writing—original draft. Xingpeng Wang: Methodology, Software, Investigation. Lei Zhu: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing—review & editing. All authors reviewed the manuscript.

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Data availability

The data and analytic code necessary to reproduce the analyses presented here are available from the corresponding author (judy1981_81@hotmail.com) upon reasonable request, as are the materials necessary to attempt to replicate the findings.

Declarations

Ethics approval and consent to participate

The study was approved by the ethics committee of Fudan University [FDU-SSDPP-IRB-2020-2-002] and with the 1964 Helsinki Declaration. Every participant gave written informed consent to participate in the study and received monetary compensation.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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